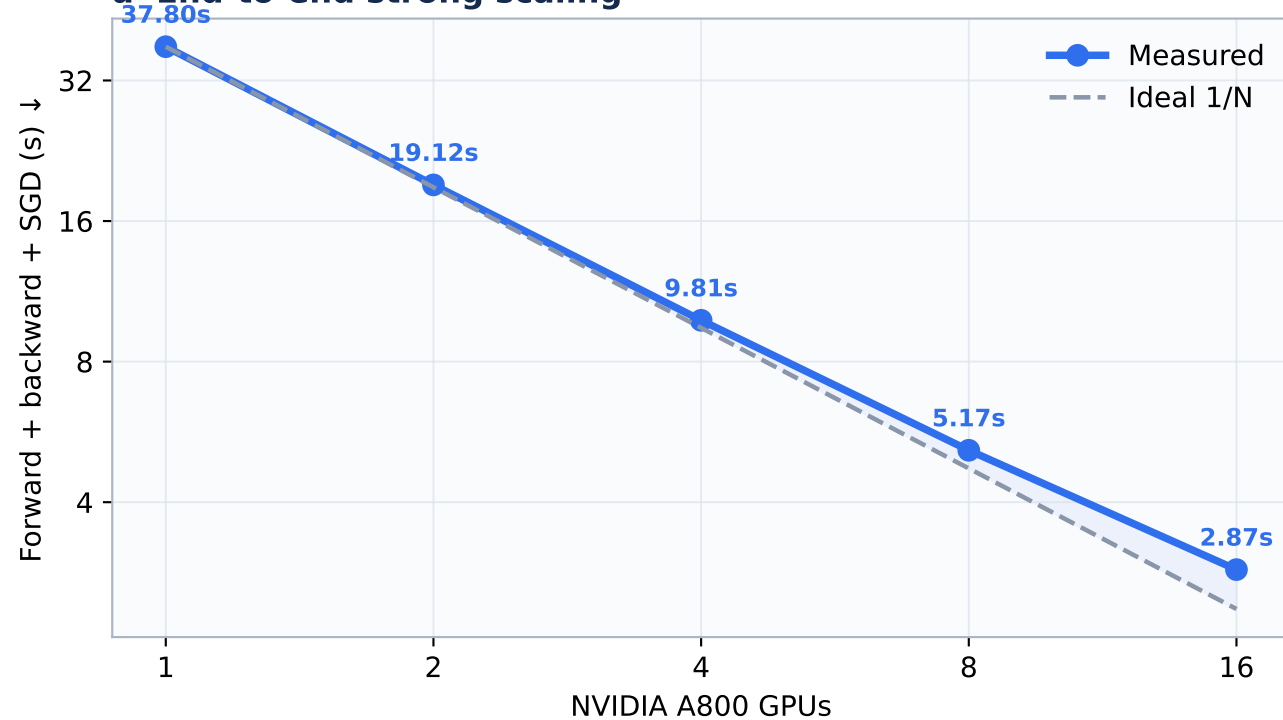
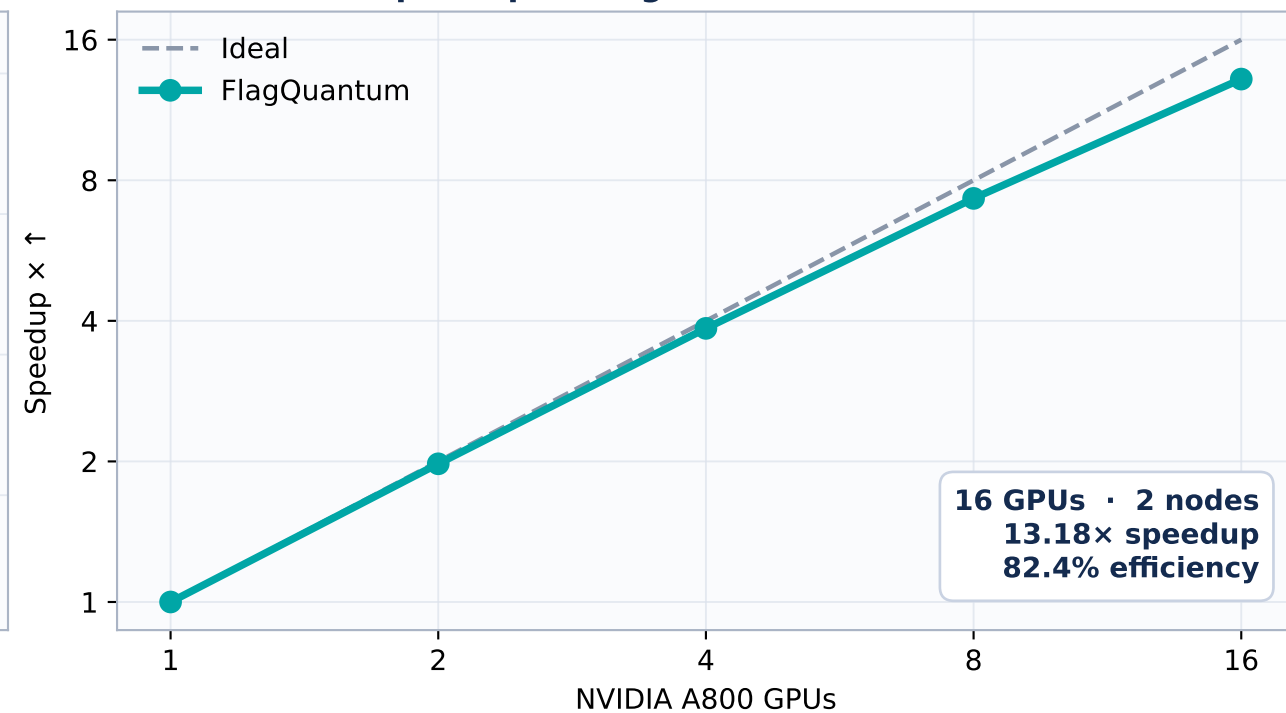


FlagQuantum · differentiable tensor-network training at 40 qubits

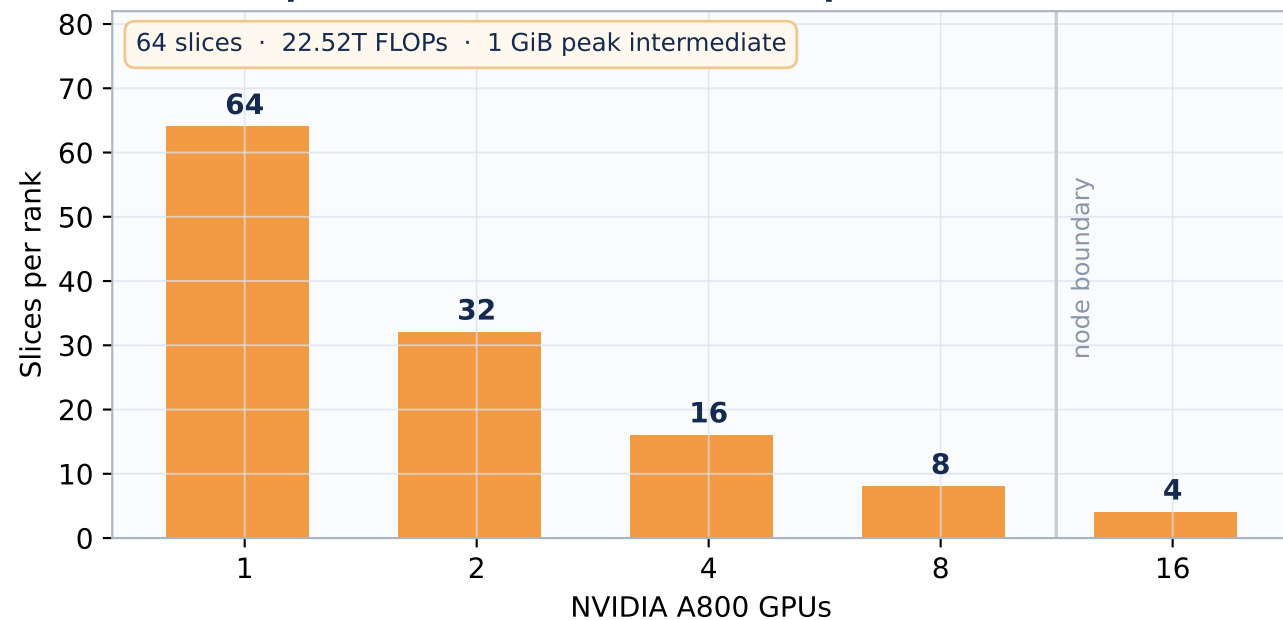
a End-to-end strong scaling



b Near-linear speedup through two nodes



c Frozen path, balanced slice ownership



d Verified differentiable TN system

40q
qubits · 5×8 grid
statevector-intractable

F+B+SGD
training semantics
owner-sharded gradients

RDMA
cross-node transport
NCCL NET/IB · RoCE

1-16 GPU
numerical stability
identical value & $\|\nabla\|_2$

Fixed 5×8 circuit · 4 entangling cycles · end-to-end complex128 · 64-slice cotengra path · explicit reverse · A800